MILITARY SPECIFICATION

SEWAGE DISPOSAL SYSTEM, EVAPORATIVE TYPE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an evaporative type sewage disposal system for shipboard installation. The sewage disposal system shall concentrate and provide storage for waste output prior to discharging to a shore receiving facility or at sea. The system shall be of the controlled-volume flush/evaporative type.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-N-281 - Nickel-Copper-Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-D-1000 - Drawings, Engineering and Associated Lists.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.
MIL-B-13239 - Barrier Material, Waterproofed, Flexible, All Temperatures.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-M-15071 - Manuals, Technical: Equipment and Systems Content Requirements for.
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-P-16789 - Packaging of Pumps, Including Prime Movers and Associated Repair Parts.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-B-24480 - Bronze, Nickel-Aluminum Castings, For Sea Water Service.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-471 - Maintainability Demonstration.
MIL-STD-769 - Thermal Insulation Requirements For Machinery and Piping.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, Center Building, SEC 6124, Prince George's Center, Hyattsville, Maryland 20782 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY (Continued)

MIL-STD-777 - Schedule of Piping, Valves, Fittings and Associated Piping Components for Surface Ships.
MIL-STD-781 - Reliability Tests: Exponential Distribution.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1629 - Procedures for Performing a Failure Mode and Effect Analysis for Shipboard Equipment.

DRAWINGS

MILITARY

NAVSHIPS 810-1385772 - Mounts, Shock, for Vitreous China Urinals, Back Mounted.
NAVSHIPS 810-1385774 - Mounts, Shock for Vitreous China Water Closets, Back Mounted.

PUBLICATION

MILITARY

NAVSHIPS 0900-LP-001-7000 - Fabrication and Inspection of Brazed Piping Systems.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INCORPORATED, AGENT
National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P Street N.W., Washington, D.C. 20036.)

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS (IEEE)
Standard 45 - Recommended Practice for Electric Installations on Shipboard.

(Application for copies should be addressed to the Institute of Electrical and Electronic Engineers, Inc., 345 East 47th Street, New York, N.Y. 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.3).

3.2 Material. Galvanic corrosion shall be considered in the selection of materials. Silver-brazed joints shall conform to the requirements of NAVSHIPS 0900-LP-001-7000.

3.3 General design. The system shall be capable of collecting and transferring the sewage to an evaporator tank against a 50-foot head with a small quantity of flushing water. A small quantity of flushing water is defined as no more than 1 pint for urinal and no more than 3 pints for water closets. Sewage, as defined in this specification, refers to human body wastes. The system shall be designed to operate utilizing sea water for flushing. The system shall consist of the following components:

(a) Controlled-volume flush water closet.
(b) Controlled-volume flush urinal.
(c) Macerator/transfer pump.

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(d) Evaporator tank.
(e) Discharge pump.
(f) Automatic controls.
(g) Vapor treatment system.

3.3.1 Controlled-volume flush water closet. The controlled-volume flush water closet shall flush the human wastes to a macerator/transfer pump for subsequent transfer to an evaporator tank. The flushing and waste transfer shall be accomplished either manually, by use of a foot pedal, or by an electronic actuator utilizing not more than 3 pints of sea water to transport wastes and clean the fixture. The water closet shall be of vitreous china, shall weigh a maximum of 120 pounds, and shall be shock-mounted as specified in 3.3.9.3.

3.3.2 Controlled-volume flush urinal. The controlled-volume flush urinal shall flush with not more than 1 pint of sea water. The urinal shall be of vitreous china, shall weigh a maximum of 75 pounds and shall be shock-mounted in accordance with 3.3.9.3.

3.3.3 Macerator/transfer pumps. The macerator/transfer pumps shall have a capacity of 1-1/2 horsepower (hp) and shall comminute and transfer the waste materials from the water closets and urinals through the soil lines to the evaporator tank. The pumps shall be capable of being mounted above or below deck in soil drain lines, comminuting all types of sanitary wastes and deleterious materials which may enter a water closet (such as shop rags, razor blades, paper towels, screws, nails, keys) and pumping these wastes up to heads of not less than 50 feet at rates of not less than 25 gallons per minute (gal/min). The cutter blades shall be constructed of a Haynes Stellite Alloy No. 3 or No. 6 casting, or equal, and the pump shaft shall be nickel-copper alloy in accordance with QQ-N-281. The pump casing shall be constructed of nickel-aluminum bronze in accordance with MIL-B-24480. Other parts shall be nickel-copper alloy in accordance with QQ-N-281. The macerator/transfer pump shall weigh a maximum of 120 pounds.

3.3.4 Evaporator tank. The evaporator tank shall process sewage by evaporating a minimum of 30 gallons of liquid per day. As the sewage is concentrated into a sludge which accumulates to a preset level, automatic controls shall signal that evaporator servicing is required. Servicing shall consist of draining the evaporator by means of the discharge pump and subsequent rinsing with (90 pounds per square inch (lb/in^2)) of fire main flush water. The tank shall be constructed of stainless steel and coated on the inside surfaces, including the exterior surfaces of the liquid level controls, with a 4 to 5 mil thickness of fluorinated ethylene propylene. A gasketed 10-inch diameter see-through access port shall be provided for performing regular maintenance and examination of the tank's interior. External surfaces shall be insulated in accordance with MIL-STD-769 for safety of personnel. The lower section of the tank shall be shaped to facilitate the pump-out process. The tank shall weigh a maximum of 750 pounds empty.

3.3.5 Discharge pump. A discharge pump shall be provided which shall be capable of pumping a sludge containing up to 20 percent by weight of solids from the evaporator tank up to heads of not less than 50 feet at rates of no less than 25 gal/min. Materials shall conform to commercial marine standards. The discharge pump and motor shall weigh a maximum of 40 pounds.

3.3.6 Automatic controls and indicators. Automatic controls and indicators shall be provided to perform the operations specified in 3.3.6.1 through 3.3.6.3. An enclosure shall be provided in which electrical indicators and electrical manual controls shall be housed.

3.3.6.1 Pump controls. Upon receipt of a signal indicating that a commode or urinal has been flushed, the controls shall start and operate the macerator/transfer pump for a period of approximately 10 seconds. A liquid level control shall be provided to prevent the operation of the pumps when the evaporator is full.

3.3.6.2 Heater controls. The tank heater shall be activated by a liquid level float type sensor when the sewage in the tank reaches a predetermined height or volume. As the water boils off and the liquid level decreases, the heater shall be deactivated. Sensors shall also be provided to prevent damage to the heaters from overheating.

3.3.6.3 Servicing controls. Controls and indicators shall be provided to signal that the sludge concentration in the evaporator tank has accumulated to a preset level and that evaporator servicing is required.

3.3.7 Vapor treatment system. A low temperature (500°F) catalyst bed shall be provided to destroy odors in steam effluent. Weight shall be limited to 100 pounds.

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3.3.8 Safety. Rotating parts, electrical parts, and parts subject to high temperatures that are of such a nature, or so located, as to become a hazard to personnel, shall be insulated, fully-enclosed, or properly guarded.

3.3.9 Environmental design requirements.

3.3.9.1 Evaporator vapors. The evaporator vapors shall not contain objectionable odors or hazardous toxic gases and shall include a vapor treatment system with a cayalyst bed.

3.3.9.2 Fungus. The system components shall not support fungus growth.

3.3.9.3 Shock. The water closet and urinal assemblies shall be shock-mounted in accordance with Drawings 810-1385772 and 810-1385774, modified as required for variations in shape as well as number of mounting holes to match the selected fixtures.

3.3.9.4 Temperature. Components shall be designed to operate at ambient temperatures between 32°F and 120°F with the exception of the evaporator internal surfaces.

3.3.9.5 Trim, list, pitch, and roll. The system shall operate under the following conditions of trim, list, pitch, and roll:

(a) When the ship is permanently trimmed down by the bow or stern as much as 5 degrees from the normal horizontal plane.
(b) When the ship is permanently listed up to 15 degrees from either side of the vertical.
(c) When the ship is pitching 10 degrees up and down from its normal horizontal plane, in a full period of 5 seconds.
(d) When the ship is rolling up to 45 degrees either side of the vertical, in a full period of 10 seconds.

3.3.10 Electrical requirements. The system shall be designed to operate from a 3-phase, 440-volt, alternating current (a.c.), 60 hertz (Hz) or 400 Hz power source as specified (see 6.1.1). Average daily operating requirements shall be within the following power constraints:

Evaporator - 200 watts/manday
Pumps (each) - 2.3 kilowatts (kW) for 10 second flush
Controls, etc. - 0.5 kW for 24 hours

3.3.10.1 Motors. Motors for driving pumps shall conform to IEEE 45 and the following:

(a) Horsepower - As required.
(b) Enclosure - Totally-enclosed, fan-cooled.
(c) Insulation - Class B, F, or H.
(d) Ambient temperature - 50°C.
(e) Motor temperature rise - Not to exceed 130°C total (ambient plus rise).

3.3.10.2 Piping and valves. Valves, fittings, flanges, and bolting shall conform to MIL-STD-777 for sea water services.

3.3.10.3 Starting controls and wiring. Starting controls and wiring shall be in accordance with IEEE 45.

3.4 Reliability and maintainability requirements.

3.4.1 Reliability. The specified mean-time-between-failure (MTBF) shall be 500 hours.

3.4.1.1 Definition of failure. The definition of failure for the purpose of determining the achievement of the reliability shall be any malfunction of the components specified in 3.3.1 through 3.3.7 that requires shutdown of the system or diversion of sewage overboard.

3.4.2 Maintainability. Ninety-five percent of all possible fault corrective actions shall not individually exceed 5 hours.

3.5 Failure mode and effects analysis. The contractor shall perform a failure mode and effects analysis (FMEA) in accordance with MIL-STD-1629 and shall prepare a FMEA report in accordance with the data ordering document included in the contract or order (see 6.1.2). The analysis shall be conducted to the functional subassembly level specified in MIL-STD-1629, except for those failure modes with a level of severity of 4. For these failure modes, the

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analysis shall be extended to the individually replaceable item level. The FMEA shall identify the design provisions that have been incorporated to inhibit or limit the frequency of occurrence of failure modes with a level of severity of 3 or 4.

3.6 Painting. External surfaces shall be thoroughly cleaned and painted in accordance with commercial marine standards. Components fabricated of stainless steel or vitreous china do not require painting.

3.7 Identification plates. Each component (see 3.3) shall be equipped with identification plates conforming to types A or B of MIL-P-15024 and MIL-P-15024/5, including warnings and cautions suitably located describing any special or important procedure to be followed in operating and servicing the system. Identification plates of plastic or aluminum are not acceptable.

3.8 Onboard repair parts. Onboard repair parts shall be furnished in accordance with MIL-P-15137 (see 6.2).

3.9 Technical data. The contractor shall prepare technical data in accordance with the data ordering documents included in the contract or order (see 6.1.2) and as specified in 3.9.1 through 3.9.2.1.

3.9.1 Engineering drawings. In addition to the drawing content required by the data ordering document (see 6.1.2), the unique features specified in 3.9.1.1 shall be included.

3.9.1.1 Sectional assembly drawings. Sectional assembly drawings shall contain the following:

(a) Bill of material, listing materials, material specifications, quantities, and manufacturer's part number.
(b) Dimensions and location of all tube or pipe connections, foundations, bolting, etc., pertinent to installation or maintenance.
(c) Schematic diagram, showing relative location of valves, fittings, controls, and components furnished with the system.
(d) Motor drawings shall show overall dimensions, shaft dimensions, motor characteristics, winding data, schematic diagram, list of repair parts, and insulation material and detail of bearing arrangement.

3.9.2 Technical manuals. In addition to the general requirements covered by the data ordering document (see 6.1.2), the unique features specified in 3.9.2.1 shall be included.

3.9.2.1 Technical manuals shall provide instructions on the installation, operation, and maintenance of the system. A detailed wiring diagram and parts list shall be included. Quantity of manuals shall be as specified in the contract or order.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system program plan. The contractor shall provide and maintain an inspection system program plan in accordance with the data ordering document included in the contract or order (see 6.1.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the tests specified in 4.5.1 through 4.5.8.

4.4 Quality conformance inspection. Quality conformance inspection shall be conducted on each system offered for delivery. The systems shall be subjected to the tests specified in 4.5.5 and 4.5.6.

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4.5 Test methods.

4.5.1 Weight. Weights shall not exceed those specified in 3.3.1 through 3.3.7.

4.5.2 Evaporator leak test. The evaporator shall not leak when filled with water and heated to 212°F.

4.5.3 Performance test. The performance test shall be accomplished at a U.S. Navy Laboratory or facility in a waste loaded condition. Subsequent to first system successfully completing the performance test, other systems of the identical configuration and type are not required to be performance tested.

4.5.4 Shock test. The system shall be subjected to the grade B shock tests as specified in MIL-S-901.

4.5.5 No-load test. The system shall be assembled and operated at the rated voltage and no load condition to ascertain that components are properly assembled and in working condition.

4.5.6 Hydrostatic pressure test. The system shall be hydrostatically tested for a period of 1 hour. The evaporator tank steam jacketed relief valve (gage pressure of 30 lb/in^2) shall be pressure tested. Pump discharge piping shall be tested at 50 lb/in^2.

4.5.7 Reliability demonstration. The contractor shall perform a reliability demonstration in accordance with MIL-STD-781 and shall prepare a test plan in accordance with the data ordering document included in the contract or order (see 6.1.2). The test shall be conducted using the following provisions:

(a) The specified MTBF (see 3.4.1).
(b) The test plan shall be number III.
(c) The definition of failure shall be as specified in 3.4.1.1.
(d) In the event of inability to reach an "accept" decision, the contractor shall perform an analysis to determine the cause, perform corrective action, and propose appropriate modifications to the design. Verification of success of any modifications made shall be accomplished by a period of retest to be determined by the procuring activity.

4.5.8 Maintainability demonstration. The contractor shall perform a maintainability demonstration in accordance with MIL-STD-471 and shall prepare a test plan in accordance with the data ordering document included in the contract or order (see 6.1.2). A minimum of 20 maintenance tasks shall be performed to demonstrate that the requirements of 3.4.2 are met. Proposed maintenance tasks shall be included as part of the maintainability demonstration test plan. Actual task selection shall be made by the procuring activity at the time of test.

4.6 All failures, deficiencies, and discrepancies revealed during the performance of the tests and the corrective measures taken shall be recorded and fully documented in the applicable test reports (see 4.7). After correction of deficiencies, tests shall be repeated to the extent necessary to ascertain acceptability of the modified system. Major failures indicative of the design deficiency (as distinguished from error or faulty workmanship) shall be reported to the procuring activity (Government or shipbuilder), before a correction is made.

4.7 Test procedures and test reports. The contractor shall prepare test procedures and test reports in accordance with the data ordering documents included in the contract or order (see 6.1.2) and as specified in 4.7.1.

4.7.1 Each test shall be fully documented by a prescribed test procedure and a written record of the performance and findings of each test. A compilation of the documentation of all tests performed on each system shall be furnished in duplicate to the cognizant Government quality assurance office and to the procuring activity (Government or shipbuilder).

4.8 Inspection of preparation for delivery. Preservation-packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

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5.1 General requirements.

5.1.1 Packaging technical data. Data of the contractor's proposed preservation-packaging, packing, and marking concept(s), including packaging requirements code(s), shall be furnished in accordance with the data ordering documents included in the contract or order (see 6.1.2).

5.2 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.1.1).

5.2.1 Level A. Sewage disposal systems shall be preserved-packaged as follows:

(a) Water closets, urinals, tanks, and reservoirs shall be individually preserved-packaged to meet the requirements of MIL-P-116, method III, by the application of cushioning, anchoring, bracing, and blocking specified in 5.6, utilizing the containers listed in 5.3.1 for the level specified (see 6.1.1). All openings, including piping units and the like, shall be plugged, capped, or sealed to prevent entrance of foreign materials.
(b) Pumps in accordance with MIL-P-16789.
(c) Electrical items, including attached junction boxes, in accordance with MIL-E-17555.
(d) Mechanical accessories in accordance with MIL-R-196 or MIL-P-16789, as applicable.

5.2.2 Level C. Sewage disposal systems shall be cleaned, dried, preserved-packaged to afford protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity for immediate use. The contractor's normal preservation-packaging methods may be utilized when such meets the requirements of the level.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.1.1).

5.3.1 Levels A and B.

5.3.1.1 System parts. System parts, except replacement parts (see 5.4), of one complete system shall be packed in containers as specified in table I for the level of packaging required. Selection of the container shall be at the contractor's option.

TABLE I. Container selection.

Specification[1/]	Container[3/]	Application (style, type, or class)	
		Level A	Level B
PPP-B-636	Box - fiberboard	Weather resistant	Domestic
PPP-B-640	Box - fiberboard corrugated triple-wall	Weather resistant	Non-weather resistant
PPP-B-585[2/]	Box - wood, wirebound	Class 3	Class 1
PPP-B-601[2/]	Box - wood, cleated-plywood	Overseas type	Domestic type
PPP-B-621[2/]	Box - wood, nailed and lock corner	Class 2 overseas	Class 1 domestic

1/ Maximum gross weight, container plus contents, shall not exceed the applicable container specification requirements for the style, type, or class container selected for the level of packing specified in the contract or order.

2/ For weights under 100 pounds, container may be used; however, lighter weight, minimum cube and less expensive fiberboard containers should be given consideration.

3/ Containers shall be closed, strapped, or banded in accordance with the applicable container specification or appendix thereto with method V closure for PPP-B-636, level A and method I closure for PPP-B-636, level B.

Consolidation of packaged equipment into a minimum of shipping containers is desirable, however, where minimum cube and tare weight could be adversely affected, large units such as the evaporator tank should be packed separately.

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5.3.1.2 Container caseliner and reinforcement.

5.3.1.2.1 Level A. When specified (see 6.1.1), shipping containers shall have caseliners conforming to MIL-L-10547. Barrier material conforming to MIL-B-13239 may be used to fabricate the caseliner. Caseliners shall be closed and sealed in accordance with MIL-L-10547. Caseliners for fiberboard containers may be omitted provided the containers are waterproofed with tape in accordance with the appendix to the applicable box specification (see table I, footnote 3 for PPP-B-636 boxes). In lieu of steel strapping, fiberboard boxes shall be reinforced with pressure-sensitive, reinforced filament tape and applied in accordance with the appendix to the box specification.

5.3.2 Level C. Sewage disposal systems, preserved-packaged as specified (see 6.1.1), shall be packed for shipment in a manner acceptable to the carrier, at the lowest rate, and which will afford protection against physical and mechanical damage during shipment from the supply source and insure safe delivery to the first using activity for immediate installation. The shipping containers and method of packing shall conform to the Uniform Freight Classification and the National Motor Freight Classification Rules or other carrier rules as applicable to the mode of transportation.

5.4 Repair parts. Repair parts shall be preserved-packaged, packed, and marked in accordance with MIL-R-196, MIL-P-16789, or MIL-E-17555, as applicable for the levels specified as follows:

Destination	Preservation-packaging Level	Packing Level
Stock	A	B
Shipboard (onboard)	A	C
Immediate use	C	C

5.5 Marking. In addition to any special marking required by the contract or order or herein, all interior (unit and intermediate) packages, shipping containers, and consolidated loads shall be marked in accordance with MIL-STD-129.

5.6 Cushioning, anchoring, blocking, and bracing. Cushioning, anchoring, blocking, and bracing shall be in accordance with MIL-STD-1186 and the applicable container specification or appendix thereto.

5.7 Technical manuals. Technical manuals, which accompany shipments shall be packaged in transparent, waterproof plastic bags, minimum 4 mil thick. Bag closure shall be by heat sealing. The shipping container housing the manual shall be marked "MANUAL(S) ENCLOSED". Technical manuals when shipped in bulk quantities shall be prepared for shipment in accordance with MIL-M-15071.

5.8 Drawings. Drawings shall be prepared for shipment in accordance with MIL-D-1000 for the level specified (see 6.1.1).

6. NOTES

6.1 Ordering data.

6.1.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of water closets and urinals to be connected to the system.
(c) Class of ship and crew size for evaporator service.
(d) Frequency, 60 or 400 Hz (see 3.3.10).
(e) Levels of preservation-packaging, packing, and marking required (see 5.2, 5.3, 5.3.2 and 5.8).
(f) When caseliners are required (see 5.3.1.2.1).

6.1.2 Data requirements. When this specification is used in a procurement which invokes the provision of the "Requirements for Data" of the Armed Services Procurement Regulations (ASPR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the ASPR are not invoked in a procurement, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

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Paragraph	Data requirement	Applicable DID	Option
3.5	Failure mode and effect analysis report	DI-R-2115	----
3.9.1	Drawings, engineering, and associated lists	UDI-E-23174	Categories B, D, F, G, and H, form 3, type II
3.9.2	Technical manuals	DI-M-2051	Type I of MIL-M-15071
4.1.1	Inspection system program plan	DI-R-4803	----
4.5.7	Reliability test plan	DI-R-2124	----
4.5.8	Maintainability demonstration plan	DI-R-2129	----
4.7	Test procedures	UDI-T-23649	----
4.7	First article test reports	UDI-T-23790	----
5.1.1	Preservation-packaging, packing, and marking (PPP & M), drawing	UDI-E-23123	----
5.1.1	Packaging and transportation support data	UDI-P-23508	----

(*Copies of data item descriptions* required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.1.2.1 The data requirements of 6.1.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the procuring/purchasing activity upon certification by the offeror that *identical data were submitted* by the offeror and accepted by the Government under a previous contract for identical item procured to this specification. This does not apply to specific data which may be required for each procurement regardless of whether an identical item has been supplied previously (for example, test reports).

6.2 Management control system document. The following management control system document should be included on DD Form 1660:

(a) MIL-P-15137 (see 3.8)

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH

(Project 4630-N005)

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